

Detecting double H.266/VVC compression with the same coding parameters

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ABSTRACT

The detection of video double compression with the same coding parameters is a difficult problem in video forensics, since the recompression traces are extremely slight in this case. Although a series of methods have been proposed, detection of Versatile Video Coding (H.266/VVC) double compression with the same coding parameters is rarely reported. To address this issue, we propose a novel algorithm for the detection of H.266/VVC double compression with the same coding parameters in this paper. We first analyze the generation of H.266/VVC double compression traces, and explore the variation of coding modes caused by multiple compressions. Then we define Minimum Unit Mapping (MUM) and Subunit Prediction Mapping (SPM) to facilitate feature construction. The strength of the variation in the number of coding unit (CU) in each compression is calculated to form CU partition modes based feature set. We also use the consistency ratios of adjacent prediction mode pairs in the horizontal, vertical, major diagonal, and minor diagonal directions after multiple compressions to obtain prediction modes based feature set. These feature sets are further concatenated into a fusion feature for detection. With the thorough comparison to the state-of-the-art algorithms, it is verified that the proposed algorithm outperforms other current works in the field. Besides, the proposed method is robust against various encoding configurations.

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1. Introduction

The emergence of video editing tools has brought great convenience to our daily life, however, it also has a negative impact on the authenticity and originality of digital video [1–4]. Therefore, it is worthwhile to apply passive video forensics method to detect video manipulation and editing.

As a kind of video editing, video double compression can recompress a video into any specific quality. The detection of video double compression has received more and more attention in recent years [5,6]. For example, in 2009, Wang et al. [7] pointed out that the distribution of quantized transformation coefficients would change if a video undergoes double compression. They then utilized this feature for media forensics. Similarly, many scholars tried

to explore the statistical features in coding domain to detect double compressed video. In [8], Chen et al. introduced the first digit distribution of non-zero quantized Alternating Current (AC) coefficients for the detection. Jiang et al. [9] utilized the Markov properties of transformation coefficients in four different directions to construct fusion feature. Promising performance is shown in the case of different compression qualities. More works [10–12] using the distributions of Discrete Cosine Transform (DCT) coefficients have also been proposed to tackle the issue. Since video double compression can also lead to the variation of the macroblock prediction types, Vázquez-Padín et al. [13] proposed an algorithm based on the variation in inter frames to detect double compressed Moving Picture Experts Group (MPEG) videos. These authors provided a deep theoretical analysis allowing us to understand the artifacts that justify the presence of the Variation of Prediction Footprint (VPF) in double compressed videos in [14] and put forward a generalized variation of prediction footprint for double compression detection. Since the prediction strategy of High Efficiency Video Coding (H.265/HEVC) differs from its predecessor to

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some extent, Xu et al. proposed the Sequence of Number of Prediction Unit of its Prediction Mode (SN-PUPM) feature for the detection of H.265/HEVC double compression [15], and achieves an Area Under Curve (AUC) of 0.9226. The inherent blocking artifacts in video coding also motivate scholars to explore traces in these artifacts for detection. In [4], He et al. proposed an adaptive post-filtering technique to measure the strength of block artifacts in the decompression domain. The strength sequences were used to detect double compression and estimate the size of the original Group of Pictures (GOP). Xu et al. [16] pointed out that the filtering mode in single compressed HEVC video is different from that of the transcoded video. They then adopted in-loop filtering to detect double compressed H.265/HEVC videos. Results proved that the algorithm outperforms the compared algorithms and has better robustness. The same authors also evaluated the impact of video motion information on double compression detection, and proposed a motion-adaptive detection algorithm in [17]. In addition, some deep learning-based double compression detection methods have also been proposed in recent years. For example, in [18], He et al. proposed a detection method for fake bitrate videos using a hybrid deep-learning network from recompression error. The authors designed a network that contains two branches with heterogeneous structures for the detection, the average detection accuracy for different bitrate combinations is above 92%. Besides, genetic Convolutional Neural Networks (CNN) is introduced in [19] to detect relocated I-frames. More statistical features like prediction residual [20,21], prediction mode [22–24], and the idempotency property [25,26] are also adopted to detect double compressed videos.

Although these algorithms work well when the parameters of double compression are inconsistent, some scholars found that when the video undergoes double compression with the same coding parameters, the detection would be more difficult. The reason is that when the same encoding parameters are used for video recompression, the difference between the coded videos is weaker than that of the recompression with different parameters, and the recompression traces are much more inconspicuous in this case. To tackle this problem, many scholars have proposed related algorithms in recent years. In [27], Chen et al. defined the concept of MacroBlock Mode (MBM) feature for the detection of MPEG double compression with the same Quantization Parameter (QP). Jiang et al. [28] made some improvements on this basis by calculating the variation rate of the unstable block, so that the algorithm can be applied to Advanced Video Coding (AVC) video. The general procedure includes multiple compression and decompression, intensities calculation, and classification. Results proved the robustness of these algorithms against various compression qualities. However, they cannot be directly applied to H.265/HEVC videos due to the upgrade of the block partitioning modes. Then, in [29], Intra Prediction Unit Prediction Mode (IPUPM) is adopted to evaluate the block stability. By calculating the variation rates of PUs, experimental results demonstrate the effectiveness of the algorithm for H.265/HEVC double compression detection.

In summary, the above mentioned works can achieve promising results under previous coding standards. However, when facing the latest Versatile Video Coding (H.266/VVC) standard, there will be a high probability of performance degradation or execution failure. It is because the most recent international video coding standard, H.266/VVC, was developed to serve an ever-growing need for improved video compression as well as to support a wider variety of today's media content and emerging applications [30]. The new coding standard has introduced a series of new coding techniques, such as multi-type tree partitioning, Adaptive Loop Filtering (ALF) and so on, which inevitably interferes with the feature extraction of existing algorithms. However, no relevant algorithm for H.266/VVC double compression detection has been proposed so far.

Therefore, it is of great significance to develop a novel algorithm for the detection of H.266/VVC double compression.

In this paper, we propose an algorithm to detect H.266/VVC double compression. Specifically, our work offers the following contributions: (1) We study the new problem of video double compression detection for H.266/VVC standard by leveraging modes analysis in the coding domain. (2) We calculate the variation of partition modes and consistency of prediction modes of coding units after consecutive compressions and decompressions to characterize the double compression traces. The variation in the number of CUs with different partition modes in each compression and the consistency ratios of adjacent prediction mode pairs in four directions are calculated to form the detection feature. (3) Minimum Unit Mapping (MUM) and Subunit Prediction Mapping (SPM) are designed to facilitate the feature construction. We use MUM to reduce the interference of quadtree or multi-type tree partition to mode analysis. SPM is adopted so that the number of adjacent prediction mode pairs in different directions can be calculated more accurately. (4) Extensive experiments are performed. A wide variety of scenarios and parameter settings are introduced to evaluate the effectiveness and robustness of the proposed algorithm comprehensively.

The remainder of this paper is organized as follows: In Section 2, new features of the H.266/VVC standard and in-depth analysis of the generation of H.266/VVC double compression traces are provided, while in Section 3 we introduce the detection algorithm. The experimental validation and comparison with existing methods are provided in Section 4. Finally, the conclusions are drawn in Section 5.

2. Preliminaries

2.1. Quadtree and multi-type tree structures in H.266/VVC

H.266/VVC still adopts block-based hybrid coding strategy like its predecessor H.265/HEVC, and the basic processing unit within a picture in H.266/VVC is the coding tree unit (CTU) [30]. For the block partition strategy, a picture can be divided into Coding Tree Units (CTUs), CTUs can be further divided into several coding units (CUs) by quadtree or multi-type tree partitioning with a size range: the Smallest CU (SCU) size of 8×8 to the Largest CU (LCU) size of 128×128 [31]. The definition of quadtree partition (QT) in H.266/VVC is similar to that of H.265/HEVC, but H.266/VVC supports larger CU sizes which allowing for a better adaptation to image content for high spatial resolutions. With a view of improving the coding efficiency, a new block partition structure named multi-type tree structure containing Binary Trees (BTs) and Ternary Trees (TTs) is introduced in H.266/VVC. Horizontal Binary Trees and Ternary Trees are defined as HBT and HTT, respectively; and Vertical Binary Trees and Ternary Trees are defined as VBT and VTT, respectively. For a block with a size of $4N \times 4N$, HBT and VBT split the block into two $4N \times 2N$ blocks and two $2N \times 4N$ blocks in the horizontal and vertical directions, respectively. TT splits the block into three sub-blocks. HTT performs block partitioning in the horizontal direction, and the sizes of the split sub-blocks are $4N \times N$, $4N \times 2N$, $4N \times N$. Those of VTT are $N \times 4N$, $2N \times 4N$, and $N \times 4N$. Due to the multi-type tree partition, the PUs in H.265/HEVC, which can take non-square sizes, are abandoned. It is because the CU sizes can be non-square in H.266/VVC. Therefore, in H.266/VVC, CU is also adopted as the unit for prediction and transformation, unless it is too large for the maximum transform length. Fig. 1 shows an example of quadtree and multi-type tree partition in H.266/VVC. Pictures from left to right and top to bottom, respectively represent QT partition, HBT partition, HTT partition, VBT partition and VTT partition.

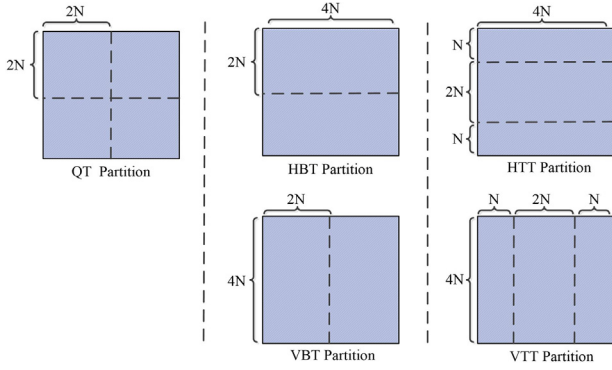


Fig. 1. Tree partition in H.266/VVC.

2.2. Intra prediction modes in H.266/VVC

Compared with H.265/HEVC, the number of angular prediction directions in H.266/VVC is increased from 33 to 65. These angular prediction directions are defined from 45 to -135 degrees in a clockwise direction for a square shape coding unit. For blocks with the width greater than height, the upper reference pixel is more likely to be used than the left reference pixel, while for blocks with the width less than height, the left reference pixel is more likely to be used. Therefore, Wide Angle Intra Prediction (WAIP) is proposed in H.266/VVC. If the shape of the coding unit is not square, the wide angle direction is used instead of the original direction. In these cases, 14 angles using prediction from the shorter side of a block are replaced by more extreme angles using prediction from the longer side, bringing the total number of angles supported in the design to 93 while the number of angular modes that can be signaled for any particular block size remains 65 [30]. The usage of WAIP would not lead to any changes in H.266/VVC intra coding, since the modes in the range of 67 to 80 are mapped as its corresponding mode in the range of 2 to 14, and the mapping of these modes in the range of -14 to -1 is also similar. Fig. 2 shows exam-

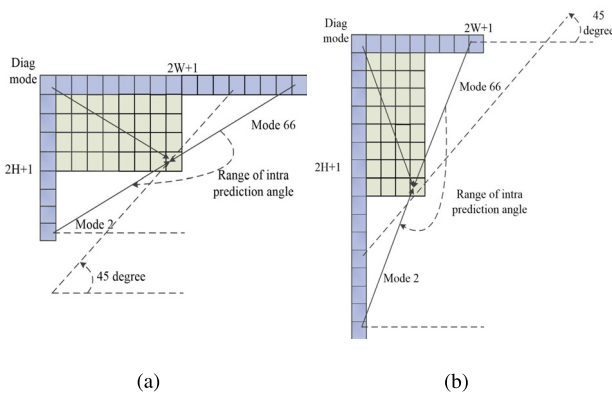


Fig. 2. Examples of reference pixels for non-square blocks: (a) Block with a size of 8×4 . (b) Block with a size of 4×8 .

Table 1

Intra prediction mode replacement in WAIP.

W/H	The replaced mode	Replace with	W/H	The replaced mode	Replace with
16	[2,3,...,15]	mode + 65	1/2	[61,62,...,66]	mode-67
8	[2,3,...,13]	mode + 65	1/4	[57,58,...,66]	mode-67
4	[2,3,...,11]	mode + 65	1/8	[55,56,...,66]	mode-67
2	[2,3,...,7]	mode + 65	1/16	[53,54,...,66]	mode-67

ples of reference pixels for non-square blocks, where the block in Fig. 2a is with a size of 8×4 , it can be observed that the number of reference pixels in the horizontal direction is 17, and the number in the vertical direction is 9. Different from the 45 to -135 degrees pixel referencing strategy, mode {2,3,...,7} is replaced with the value of the corresponding mode plus 65. Similarly, the pixel reference corresponding to the case where the block size is 4×8 in Fig. 2b is also easy to understand. For different ratios of block height and width, the detailed mode replacement strategy is shown in Table 1, W/H in the table represents the ratio of block width and height.

2.3. Analysis of the generation of H.266/VVC double compression traces

In order to explore the potentially effective features, it is necessary to analyze the generation of H.266/VVC double compression traces. As shown in Fig. 3, the original single compressed H.266/VVC video is generated in the first round compression procedure. After decoding, it undergoes the second round H.266/VVC compression and is encoded into a double compressed video. If the parameters used in these two compressions are consistent, this procedure is defined as double compression with the same parameters.

Following the coding procedure of H.266/VVC in Fig. 4, the whole procedure can be divided into block partition, block-based frame prediction, transformation and quantization, inverse transformation, in-loop filtering. We summarize the mathematical symbols of the corresponding operations in Table 2. On this basis, the transformation coefficient of single compressed video can be calculated as $\mathbf{C}^{(1)} = R(T(\mathbf{S}^{(1)} - \mathbf{P}\mathbf{S}^{(1)})/\mathbf{Q})$. Then, $\mathbf{C}^{(1)}$ undergoes the backward decoding procedure to form the reconstructed coefficient, i.e., $\mathbf{Rec}^{(1)} = RT(IT(\mathbf{C}^{(1)}/(\mathbf{Q})^{-1}))$. After in-loop filtering, the filtered

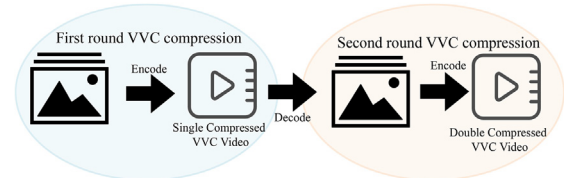


Fig. 3. Diagram of video double compression.

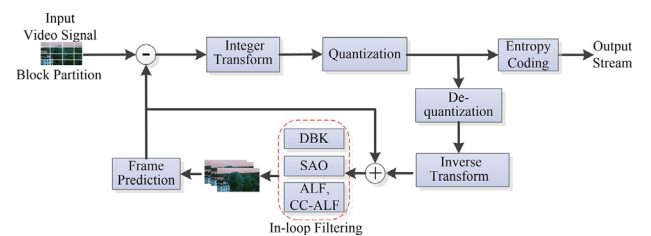


Fig. 4. The process of H.266/VVC video coding.

Table 2

Notation.

Symbol	Definition
$T(\cdot)$	Transform
$(\cdot)/\mathbf{Q}$	Quantization
n	The time of compression
$\mathbf{C}^{(n)}$	Quantized DCT/DST coefficient in the n^{th} compression
$\mathbf{S}^{(n)}$	Input Value of the source YUV video in the n^{th} compression
$\mathbf{PS}^{(n)}$	The Predicted value of $\mathbf{S}^{(1)}$ in the n^{th} compression
$R(\cdot)$	Rounding operation
$\mathbf{ReC}^{(n)}$	Reconstructed coefficient in the n^{th} compression
$RT(\cdot)$	Rounding and truncating
$IT(\cdot)$	Inverse Transform
$\Psi(\cdot)$	In-loop filter
$\mathbf{S}_{\Psi}^{(1)}$	Filtered reconstructed value

reconstructed value can be denoted by: $\mathbf{S}_{\Psi}^{(1)} = \Psi(\mathbf{PS}^{(1)} + RT(IT(\mathbf{C}^{(1)}/(\mathbf{Q})^{-1})))$, and the reconstructed frame would be sent to the buffer.

If the video undergoes double compression, the output of the first coding would be used as the input of the second coding. Then, $\mathbf{C}^{(2)}$ can be calculated as: $\mathbf{C}^{(2)} = R(T(\mathbf{S}_{\Psi}^{(1)} - \mathbf{PS}^{(2)})/\mathbf{Q})$. Further, the filtered reconstructed value of double compressed video can be represented by: $\mathbf{S}_{\Psi}^{(2)} = \Psi(\mathbf{PS}^{(2)} + RT(IT((T(\mathbf{S}_{\Psi}^{(1)} - \mathbf{PS}^{(2)})/\mathbf{Q})/\mathbf{Q}^{-1})))$.

By subtracting $\mathbf{S}_{\Psi}^{(1)}$ and $\mathbf{S}_{\Psi}^{(2)}$, the double compression traces can be formulated as: $\Delta\mathbf{DC} = \mathbf{S}_{\Psi}^{(1)} - \mathbf{S}_{\Psi}^{(2)}$. It can be observed from the derivation that $\Delta\mathbf{DC}$ is mainly caused by rounding, truncation, and filtering. On the topic of video distortion analysis, [29] presented a discussion concerning how rounding, truncation, and filtering affect the video quality degradation after multiple compression. Using statistical analysis, the authors pointed out that the quality degradation of intra-frame HEVC coding mainly comes from the integer error and truncation error in the quantization and reverse quantization procedures, and the error caused by in-loop filtering tends to be zero matrix. By exploring further, it is easy to understand that the truncation and rounding rules in H.266/VVC are almost the same as H.265/HEVC.

Therefore, we specifically analyze the effect of in-loop filtering in H.266/VVC on the quality degradation of multiple compressed video in this section. YUV sequence *in_to_tree* is selected as the test sequence, and QP is selected from {5, 25, 35, 50} for both single and double compression. We perform 5 consecutive compressions and decompressions on the test sequence by using VTM-14.0 software and calculate the Peak Signal to Noise Ratio (PSNR) between filtered and unfiltered videos. The usage of filtering is achieved by turning the filtering switch on and off in the *cfg* file. Finally, we calculated the variance and the mean of PSNR difference between filtered and unfiltered videos according to Eq. 1 and Eq. 2, where σ , M

represent the standard deviation and the mean. T, NI are the compression times and the number of intra coded frames respectively. $IP_{i,n}$ is the difference in PSNR of the n^{th} filtered and unfiltered I-frames in the i^{th} compression. Then results can be obtained as reported in Table 3. The binary codes in the table represent the states of the three filters in the loop, where the first bit represents the state of deblock filter. The second and third bits represent those of SAO filter and adaptive loop filter, respectively.

$$\sigma = \sqrt{\frac{1}{T} \times \frac{1}{NI} \sum_{i=1}^T \sum_{n=1}^{NI} (IP_{i,n} - M)^2} \quad (1)$$

$$M = \frac{1}{T} \times \frac{1}{NI} \sum_{i=1}^T \sum_{n=1}^{NI} IP_{i,n} \quad (2)$$

As can be observed from Table 3, the PSNR difference between filtered and unfiltered I-frames is maintained at around 0.326~0.451. When all three filters are turned off, the difference achieves the largest value. Anyway, no matter how the filters are set, the standard deviations remain at small values which means the quality degradation characteristic still exists in H.266/VVC double compression. Therefore, for the detection of H.266/VVC double compression with the same coding parameters, we propose a detection algorithm based on the analysis of CU partition and prediction modes.

3. The proposed algorithm

3.1. CU partition in multiple compressed video

Video multiple compression, including H.266/VVC double compression, would inevitably introduce additional traces. These traces would definitely affect the rate-distortion optimization results and have an impact on the encoded output to some extent. In this section, we analyze the variation of CU partition after H.266/VVC multiple compression to verify this point of view and to promote the exploration of reasonable feature characterization, so as to design effective detection features and provide solutions for H.266/VVC double compression detection. We select *hall_cif.yuv* and *akiyo.yuv* as the test sequences. Then we perform single H.266/VVC compression on these sequences by using VTM-14.0 coding tool. The compiled *DecoderApp.exe* file is used for decoding, and its output is further encoded into 266 files. The value of QP is set to 32 when compressing *hall_cif.yuv*, and we perform double compression on *akiyo.yuv* with QP fixed at 20. It is worth noting that the same parameters are used in the above compressions. We set the parameters *ENABLE_TRACING* and *K0149_BLOCK_STATISTICS* in *TypeDef.h* of VTM-14.0 to 1, thus enabling block statistics,

Table 3

The statistics of the PSNR difference between filtered and unfiltered I-frames.

QP = 5								QP = 25							
	000	100	010	110	001	101	011		000	100	010	110	001	101	011
M	0.522	0.478	0.483	0.417	0.440	0.402	0.414	M	0.472	0.410	0.388	0.325	0.390	0.302	0.310
σ	0.071	0.062	0.077	0.060	0.057	0.072	0.070	σ	0.072	0.068	0.072	0.078	0.080	0.074	0.074
Avg	M = 0.451, σ =0.067							Avg	M = 0.371, σ =0.074						
QP = 35								QP = 50							
	000	100	010	110	001	101	011		000	100	010	110	001	101	011
M	0.454	0.407	0.380	0.315	0.400	0.306	0.325	M	0.430	0.311	0.325	0.300	0.324	0.287	0.305
σ	0.069	0.058	0.065	0.073	0.055	0.072	0.066	σ	0.055	0.062	0.071	0.076	0.059	0.063	0.066
Avg	M = 0.370, σ =0.065							Avg	M = 0.326, σ =0.065						

which can be analysed with YUView [32]. The file name of the produced trace file and trace rule is also defined on the decoding side.

By performing three consecutive compressions and decompressions, CU partition in the luma component of the first I-frame is analyzed as shown in Fig. 5. Subfigures (a–c) and (d–f) are the results after single, double and triple H.266/VVC compression of the two sequences, while (g–i) and (j–l) are their respective zoomed version. The differences of block partition between the current block and their previous encodings are marked with red numbers. As can be seen, after H.266/VVC encoding, CUs can be squares with size $N \times N$, where $N \in \{4, 8, 16, 32, 64\}$, or a series of non-square sizes, e.g., $4 \times 8, 8 \times 16, 16 \times 32$. CUs with smaller sizes are more inclined to be utilized in the edge or complexity part of the picture, while those with larger sizes usually appear in smooth areas. After multiple compressions, the CU partition at the same position of these frames is not fixed. As shown in subfigures (g–i), compared with subfigure (g), the partition of areas marked with red numbers in (h–i) is significantly different. This phenomenon can also be observed in subfigures (j–l). The result can be attributed to the introduction of double and triple compression traces. Besides, due to the quality degradation mechanism, the changes caused by triple compression seem to be less than those of double compression. Therefore, we consider using this characteristic to design detection features for double compression detection. Previous works imply that pixel perturbation caused by these traces would introduce instabilities in CU partition and intra prediction [29,21,24]. This motivates us to evaluate the variation of CU partition and consistency of prediction modes for the detection of H.266/VVC double compression.

3.2. Definition of variation of partition modes of coding units feature

Since double compression would cause variation in CU partition, the variation of partition modes of coding units feature is constructed for the characterization of H.266/VVC double compression traces. We first perform Minimum Unit Mapping (MUM) to reduce the interference of quadtree or multi-type tree partition to mode analysis, also to facilitate the calculation of block number. MUM is a mapping from the source CU to the minimum unit. In this work, the source CU is defined as $SouCU = (Md, Lb, IaP)$, where

Md represents the partition modes of source CU, and each mode is tagged with its label (Lb), IaP is the intra prediction mode. Then, the MUM strategy can be given as: $MUM : (Md) \rightarrow (4 \times 4, Lb)$. The goal is to divided the source CUs evenly into 4×4 minimum units, and those smallest units in the same CU are labeled with the same type code. As shown in Table 4, we deliberately choose $32 \times 32, 16 \times 16, 8 \times 8, 4 \times 4, 4 \times 8, 8 \times 4, 8 \times 16, 16 \times 8, 16 \times 32, 32 \times 16, 4 \times 16, 16 \times 4, 8 \times 32, 32 \times 8, 4 \times 32, 32 \times 4$ CUs in the luma component for the calculation because of their quantities, and we label them from 1 to 16 without repetition. It is easy to understand that a 16×16 unit can be divided into 16 minimum units. According to the look-up table, the label of 16×16 partition mode is 3. Therefore, these 16 minimum units are also marked with 3. By traversing all the units, the division and mapping can be realized. Fig. 6 shows the MUM mapping result of a 32×32 block, as can be seen, the original 4×4 unit in Fig. 6(a) is labeled with 1, and the unit with a size of 8×16 is divided into 8 4×4 subunits shown in Fig. 6(b), which are labeled with 7. In this way, we can effectively measure the variation of CU partition modes after mapping, thereby reducing the interference caused by the complex CU partition modes in H.266/VVC standard.

The measurement of the strength of the variation in the number of CUs is performed on the mapped units of the input video. Firstly, the CU partition modes in intra coded frames are extracted during decoding procedure. Then, MUM is performed and the number of

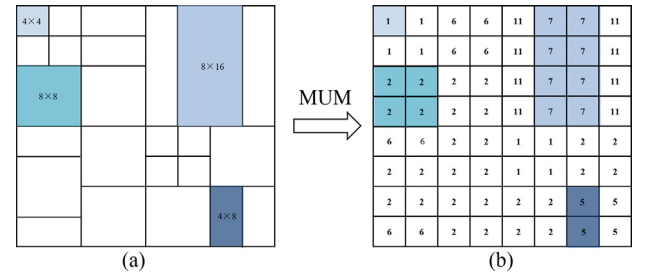


Fig. 6. (a) Example of CU partition modes in H.266/VVC and (b) Minimum unit mapping with labels in units of 4×4 .

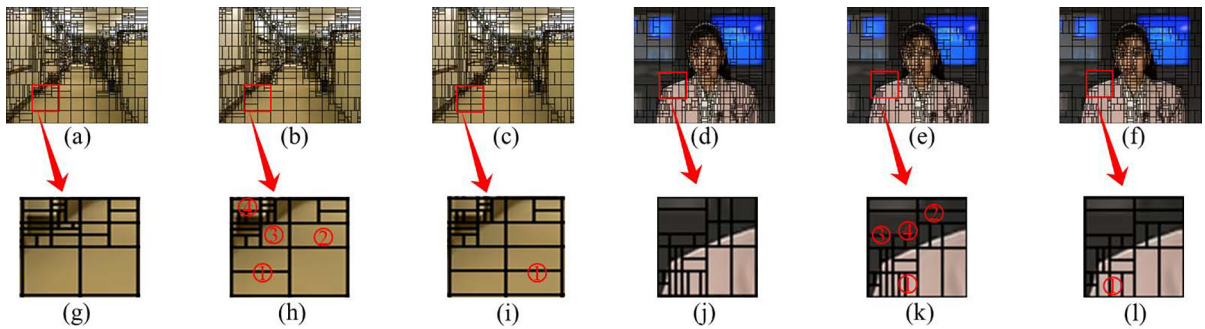


Fig. 5. CU partition of the first I-frame of video clips *hall_cif* and *akiyo*. (a–f) The partition modes of single, double, triple compressed videos. (g–l) Zoomed version of the partition modes of the 64×64 blocks marked with red boxes.

Table 4

Indexes of CU partition modes in intra frame prediction.

Mode	Label	Mode	Label	Mode	Label	Mode	Label
4×4	1	4×8	5	16×32	9	8×32	13
8×8	2	8×4	6	32×16	10	32×8	14
16×16	3	8×16	7	4×16	11	4×32	15
32×32	4	16×8	8	16×4	12	32×4	16

4×4 minimum units mapped from the block of label Lb in the i^{th} compression can be calculated as:

$$NCU_{Lb}^{n,i} = \sum_{r=1}^R \sum_{c=1}^C \rho_{c,r} \quad (3)$$

where n represents the n^{th} I-frame, $\rho_{c,r}$ in the equation is defined as:

$$\rho_{c,r} = \begin{cases} 1, & \text{If the label at coordinate}(c, r) \text{ is } Lb \\ 0, & \text{others} \end{cases} \quad (4)$$

Then, the strength of the variation in the number of CUs in the i^{th} compression can be calculated as:

$$VCU_{Lb}^i = \frac{1}{NI} \sum_{n=1}^{NI} \frac{|NCU_{Lb}^{n,i+1} - NCU_{Lb}^{n,i}|}{SMU_{Lb}^{n,i+1}}, Lb \in \{1, 2, \dots, 16\} \quad (5)$$

where NI is the total number of I-frames in the video and $SMU_{Lb}^{n,i}$ is the Sum of the number of Minimum Units mapped from the block of label Lb in the n^{th} I-frame of the i^{th} compression. $(NCU_{Lb}^{n,i+1} - NCU_{Lb}^{n,i})$ in the formula represents the difference of the number of 4×4 minimum units mapped from the block of label Lb after two consecutive compressions.

Therefore, to construct the variation of partition modes of coding units feature, we perform consecutive compressions and decompressions three times, and a 48-D feature set can be obtained:

$$\mathbf{F}_{vpar} = \{VCU_{Lb}^1, VCU_{Lb}^2, VCU_{Lb}^3\}, Lb \in \{1, 2, \dots, 16\} \quad (6)$$

3.3. Definition of consistency of prediction modes of coding units feature

To fully explore double compression traces in coding domain, we also construct the consistency of prediction modes of coding units feature for double compression detection. We first define Subunit Prediction Mapping (SPM) to facilitate the evaluation of modes consistency. In this work, the source CU is defined as $SouCU = (Md, Lb, laP)$, laP represents the intra prediction mode, which can be selected from $\{0, 1, \dots, 66\}$. Then, SPM can be denoted as: $SPM : (Md, laP) \rightarrow (4 \times 4, laP)$. The source CU is mapped into a series of 4×4 subunits, and each subunit is labeled according to the intra prediction mode of the parent unit. We first extract the intra prediction mode and partition mode of each block during the decoding procedure. Blocks with different sizes are divided evenly in units of 4×4 , and subunits belonging to the same CU share the same prediction mode. Since H.266/VVC supports 67 intra prediction modes, the mode values and labels of these subunits are assigned the prediction modes of their parent units. Fig. 7(a) is an example of intra prediction modes of CUs in a 32×32 block. We perform SPM on it and the mapping result is given in Fig. 7(b). As can be seen, an 8×8 block can be divided into 4 4×4 subunits, and their prediction modes after division are consistent with that of the parent unit in Fig. 7 (a). By adopting this mapping method, the number of adjacent prediction mode pairs in different directions can be calculated more accurately.

After SPM procedure, we calculate the consistency ratios of adjacent prediction mode pairs in the horizontal, vertical, major diagonal, and minor diagonal directions after multiple compressions for feature construction. Let us define Prediction Mapping Matrix in the n^{th} I-frame of the i^{th} compression as $\mathbf{PMM}^{n,i} = (m_{r,c}^{n,i})_{R \times C}$, where $m_{r,c}^{n,i}$ is the intra prediction mode of 4×4 subunit located at (r, c) . R and C indicate the number of rows and columns after mapping respectively. We then calculate the

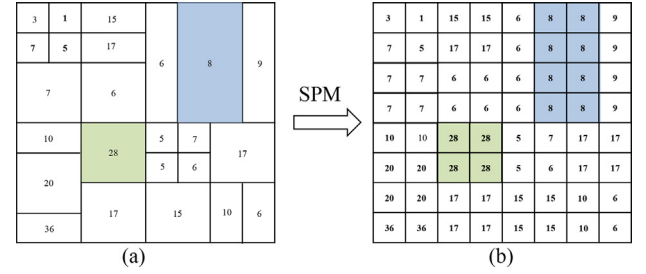


Fig. 7. (a) Example of intra prediction modes for blocks of CUs in a 32×32 block and (b) Subunit prediction mapping with labels in units of 4×4 .

number of mode pairs of two adjacent blocks with a step of 1. For the calculation in the horizontal direction, the number of adjacent prediction mode pairs combined as $(mode_1, mode_2)$ is calculated as:

$$MP(mode_1, mode_2) = \sum_{r=1}^R \sum_{c=1}^C \partial_{c,r} \quad (7)$$

$\partial_{c,r}$ in the equation is defined as:

$$\partial_{c,r} = \begin{cases} 1, & (m_{r,c}^{n,i} = mode_1) \wedge (m_{r,c+1}^{n,i} = mode_2) \\ 0, & \text{others} \end{cases} \quad (8)$$

only if the mode pairs is combined as $(mode_1, mode_2)$, the decision factor $\partial_{c,r}$ would be set to 1,

Since H.266/VVC supports 67 intra prediction modes, we construct Prediction Co-occurrence Matrix (PCM) of size 67×67 to measure the co-occurrence of these modes. The prediction co-occurrence matrix in the horizontal direction of the n^{th} I-frame of the i^{th} compression is defined as:

$$\mathbf{PCM}_{hod}^{n,i} = (\lambda_{m_1, m_2}^{n,i})_{67 \times 67} \quad (9)$$

where $m_1, m_2 \in [1, 67]$, $\lambda_{m_1, m_2}^{n,i} = MP(m_1 - 1, m_2 - 1)$.

After PCM construction, the consistency ratio is presented as:

$$CR_{hod}^{n,i} = \frac{\sum_{m_1=1}^{67} \sum_{m_2=1}^{67} m_1 \times m_2 \times \lambda_{m_1, m_2}^{n,i} - \xi_{m_1} \times \xi_{m_2}}{\omega_{m_1} \times \omega_{m_2}} \quad (10)$$

where:

$$\xi_{\tau} = \sum_{m_1=1}^{67} \sum_{m_2=1}^{67} \tau \times \lambda_{m_1, m_2}^{n,i}, \tau \in \{m_1, m_2\} \quad (11)$$

$$\omega_{\tau}^2 = \sum_{m_1=1}^{67} \sum_{m_2=1}^{67} (\tau - \xi_{\tau})^2 \times \lambda_{m_1, m_2}^{n,i}, \tau \in \{m_1, m_2\} \quad (12)$$

Therefore, the Average Consistency Ratio of the i^{th} compressed input video in the horizontal direction can be calculated as follows:

$$ACR_{hod}^i = \frac{1}{NI} \sum_{n=1}^{NI} CR_{hod}^{n,i} \quad (13)$$

where NI is the total number of I-frames.

By combining the ratios in four directions, the average consistency ratios of coding units of the i^{th} compressed input video can be formulated as:

$$\mathbf{VACR}^i = \{ACR_{hod}^i, ACR_{ved}^i, ACR_{mad}^i, ACR_{mid}^i\} \quad (14)$$

where $ACR_{ved}^i, ACR_{mad}^i, ACR_{mid}^i$ are the average consistency ratios in the vertical, major diagonal, and minor diagonal directions respectively. By performing consecutive decoding and recompression on

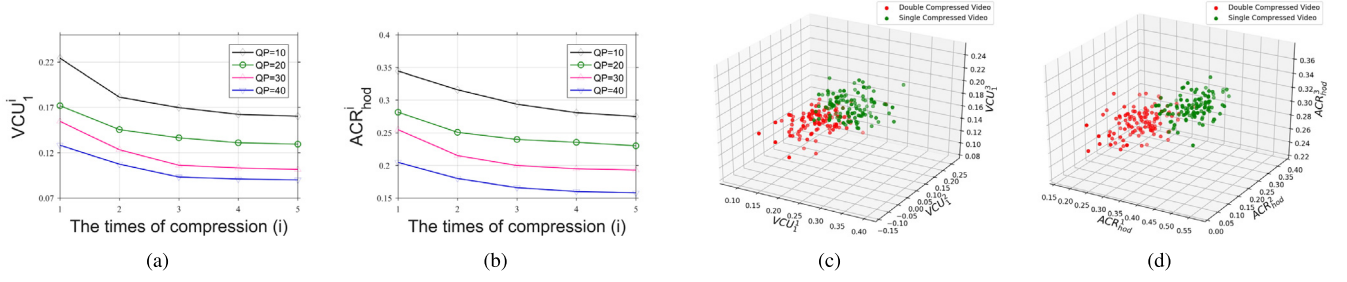


Fig. 8. (a) The variation of VCU_1^i with the compression time. (b) The variation of ACR_{hod}^i with the compression time. (c) 3-dimensional distributions of VCU_1^i when QP is fixed at 10. (d) 3-dimensional distributions of ACR_{hod}^i when QP is fixed at 10.

the input video and following the same step to calculate the consistency ratios, the related detection features denoted by $\mathbf{VACR}^{i+1}, \mathbf{VACR}^{i+2}$ can be obtained.

Combining the above terms, for the detection of H.266/VVC double compression, the consistency of prediction modes of coding units feature set can be denoted by:

$$\mathbf{F}_{cpre} = \{\mathbf{VACR}^1, \mathbf{VACR}^2, \mathbf{VACR}^3\} \quad (15)$$

Further, the 60-D fusion feature vector can be described as follows:

$$\mathbf{F}_{fusion} = \{\mathbf{F}_{vpar}, \mathbf{F}_{cpre}\}. \quad (16)$$

By evaluating the variation of partition modes and consistency of prediction modes of coding units between single and double compressed videos after recompressing with the same parameters, the feature can well represent double compression traces.

3.4. Statistical properties of the detection feature

To observe the statistical properties of the detection feature, we extract and further calculate the feature vectors of single and double compressed H.266/VVC videos. Then, we present the property curves and 3-dimensional distributions in Fig. 8.

Fig. 8a and Fig. 8b respectively show the variation of VCU_1^i and ACR_{hod}^i with the compression time. These data are calculated and averaged over 112 video clips which are comprehensively described in Section 4.1 when QP is selected from {10, 20, 30, 40}. As can be seen, with the increase of the compression time i , both VCU_1^i and ACR_{hod}^i would gradually decrease to different degrees, and the slope of each line gets smaller as i increases. This phenomenon is mainly because the trend of quality degradation would get smaller and smaller with the increase of compression times. Then the variation of pixels and the selection of coding mode tend to be stable. Since a higher QP would cause more information loss, this phenomenon is more obvious when QP gets smaller. This finding also leads us to believe that the VCU_1^i and ACR_{hod}^i of other types of CU would also follow similar rules. Fig. 8c and Fig. 8d respectively show the 3-dimensional distributions of VCU_1^i and ACR_{hod}^i of double and single compressed video clips in our dataset when QP is fixed at 10. It can be observed that these two types of feature statistics show a certain difference in distribution. So, by using the fused feature, double and single compressed H.266/VVC videos can be distinguished.

3.5. Proposed detection method

The detailed steps of the proposed algorithm are introduced as follows:

- (1) Video multiple compression and decompression. The input YUV sequence Y_0 is first encoded into H.266/VVC video V_1 with parameters P , then V_1 is decoded into YUV sequence Y_1 , and further recompressed into H.266/VVC video V_2 . The recompression parameters are consistent with P . Repeat this continuous encoding and decoding procedures 3 times.
- (2) CU partition and prediction modes extraction. During each decoding procedure, extract the statistics of CU partition and prediction modes from I-frames of the videos. For partition modes of $32 \times 32, 16 \times 16, 8 \times 8, 4 \times 4, 4 \times 8, 8 \times 4, 8 \times 16, 16 \times 8, 16 \times 32, 32 \times 16, 16 \times 4, 8 \times 32, 32 \times 8, 4 \times 32, 32 \times 4$, we label them with numbers from 1 to 16. Prediction mode statistics for all units in I-frames are also labeled with their mode numbers.
- (3) Feature construction. Perform MUM and SPM on the decoded data and calculate the number of 4×4 minimum units mapped from the block of label Lb in the i^{th} compression. After that, the difference of the number of 4×4 minimum units with the same label after two consecutive compressions can be measured. We then obtain $VCU_{Lb}^1, VCU_{Lb}^2, VCU_{Lb}^3$ of the mapped CUs according to Eq. 5. Besides, the number of adjacent prediction mode pairs are also calculated, we construct prediction co-occurrence matrixes and calculate the average consistency ratios in four directions so as to obtain $\mathbf{VACR}^1, \mathbf{VACR}^2, \mathbf{VACR}^3$. After obtaining the above subfeatures, fuse them into the final feature vector $\mathbf{F}_{fusion}$ according to Eq. 16.
- (4) Classification. *libsvm* with Radial Basis Function (RBF) kernel is adopted as the classifier. For the training procedure, $\mathbf{F}_{fusion}$ extracted from double compressed H.266/VVC video clips are tagged as 1, and those from single compressed samples are labeled as 0. Then, 5-fold cross validation is utilized for training. We first shuffle the dataset randomly, and split the dataset into 5 subsets. For each unique subset, we take the subset as a hold out or test data set, and take the remaining groups as a training data set, *gamma*, *C*, *cache_size* being set to 0.5, 1 and 100 respectively. Then, a model is fitted on the training set and we evaluate it on the test set. By repeating each testing procedure 20 times, the average detection accuracy rate is calculated. After successively iteration, a trained SVM model can be obtained, which can be further used for H.266/VVC double compression detection.

4. Experiments and analysis

After introducing the main steps of the proposed algorithm, in this section, extensive experiments are conducted for fair comparison to other work on H.266/VVC double compression detection. Regarding the robustness against more realistic scenarios, such as Constant Quantization Parameter (CQP), Average Bit Rate

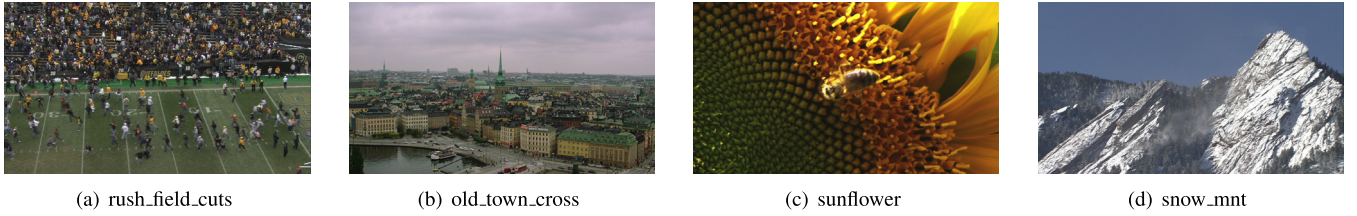


Fig. 9. Examples of YUV Sequences.

(ABR) rate control modes, different GOP sizes and spatial resolutions are verified.

4.1. Experimental setup

(1) *Dataset*: 20 1080p (1920 × 1080) raw YUV sequences[33] with diverse contents are used for datasets construction¹. Some examples are provided in Fig. 9. Since the lengths of these videos are not fixed, each sequence is split into non-overlapping YUV clips in units of 100 frames, then 112 clips can be obtained. In this work, single compressed H.266/VVC video clips are defined as negative samples. The positive samples are obtained by decoding and recompressing those singly compressed video clips with the same coding parameters. Note that, samples segmented from the same YUV clip never existed in both the training set and test set simultaneously.

(2) *Detailed Parameters*: We select Versatile Video Encoder (VVenC) as the encoder, since it is a fast and efficient “real-world” H.266/VVC encoder implementation with a series of excellent properties. *DecoderApp.exe* file compiled from VTM-14.0 software [34] using *microsoft visual studio 2015* is adopted for video decoding. GOP size is set to 10 in all experiments except for the different GOP sizes case in Section 4.5, which is selected from {10, 20, 30, 40}. For the experiment in CQP mode, QP is traversed from 1 to 51. In terms of ABR mode, the value of bitrate is selected from {2, 3, 4, 5, 6} Mbps. The option *-Preset* in Section 4.8 is respectively set to faster, fast, medium, slow, slower, while the default value is used in other experiments. For all parameter combinations, a total of 4480 single compressed and 4480 double compressed video clips are generated. Note that clips segmented from the same YUV sequence never existed in both the training set and test set simultaneously.

(3) *Evaluation Criterion*: SVM classifier is adopted for video classification and Accuracy (ACC) is used as the evaluation criterion, which can be expressed as

$$ACC = \frac{TPS + TNS}{PS + NS} \times 100\% \quad (17)$$

where TPS (TNS) is the number of true positive (negative) sample. PS (NS) represents the number of positive (negative) samples.

4.2. Performance comparison of different sub-features

In this subsection, we first evaluate the performance of each sub-feature in CQP mode. VVenC and VTM-14.0 software are used for coding and decoding respectively. By setting the option *-q* in VVenC, the value of QP can vary from 1 to 51. Then raw YUV sequences are compressed into H.266/VVC single compressed videos and these videos undergo decoding and further encoding to generate re-compressed videos. Note that, the quantization parameters used in both encodings are consistent, and during the

decoding procedure, VTM-14.0 software is used for feature extraction. We evaluate the performance on the dataset described in Section 4.1, after feature extraction and classification, we feed F_{vpar} , F_{cpre} and F_{fusion} to the classifier. Therefore, results in Table 5 can be obtained.

Results in Table 5 show that, no matter how QP varies, F_{fusion} always outperforms F_{vpar} and F_{cpre} . When QP is selected from {10, 15, 16, 20, 44, 49}, the detection accuracies of F_{fusion} reach 100%, while the highest value of the subfeatures only reach 95.71%, which indicates F_{fusion} is more efficient than F_{cpre} and F_{vpar} . In terms of average accuracy, the proposed algorithm achieves 97.33%, while F_{vpar} and F_{cpre} reach 89.47% and 88.29%, respectively. The result confirms that the detection accuracy can be effectively improved through feature fusion. The better performance of the fused feature can be attributed to the combination of the two sub-features. By leveraging traces in CU partition and prediction modes, the performance of H.266/VVC double compression detection could be significantly improved.

4.3. Comparison with some state-of-the-art algorithms

In this subsection, our algorithm is compared with Jiang et al.'s [29], Yu et al.'s [23] and Li et al.'s [35] methods in constant quantization parameter rate control scenario. For the implementation of [29], PU type and intra luma prediction mode (0–34) are extracted during the decoding process, and Intra Prediction Unit Prediction Mode (IPUPM) feature is constructed for classification. For method in [23], the frequencies of specific intra prediction modes in intra coded frames and inter prediction modes in predicted frames are calculated and further combined into a fusion feature vector for detection. For the method proposed by [35], the steganalysis feature extractor, i.e., Subtractive Pixel Adjacency Matrix (SPAM) is adopted to extract the noise residuals from video frames, and Gaussian one-class classifier is employed to achieve the detection task. By performing these algorithms in CQP mode, detection accuracy rates of these methods can be obtained, as reported in Table 6.

It can be observed that the proposed detection algorithm achieves consistently higher accuracies in most cases. Our algorithm achieves the average accuracy rate of 97.33%, which is 3.56%, 4.50%, and 5.33% higher than those of [29,23,35], respectively. When QP is selected from {10, 15, 16, 20, 44, 49}, our algorithm achieves the highest value of 100%. The better performance can be attributed to the fact that our algorithm fully exploits the double compression traces existing in the CU partition modes and intra prediction modes by increasing the exposure of the trace information through multiple encoding and decoding. [29,23,35] can achieve commendable results on AVC/HEVC double compression detection problem. However, they are not as effective as the proposed algorithm when facing H.266/VVC double compression detection. It is because H.266/VVC contains more complex CU partition modes and prediction modes, and the features proposed by these algorithms cannot fully characterize the trace information in H.266/VVC videos. It also can be observed that with the increase of QP, there's no significant decrease or increase in the detection accuracy. It is because double compression errors are inevitable

¹ The names of the YUV sequences: *aspen*, *blue_sky*, *controlled_burn*, *crowd_run*, *sunflower*, *park_joy*, *ducks_take_off*, *in_to_tree*, *old_town_cross*, *pedstrain_area*, *red_kayak*, *riverbed*, *rush_field_cuts*, *rush_hour*, *tractor*, *snow_mnt*, *speed_bag*, *touch-down_pass*, *west_wind_easy*, *station2*.

Table 5

The detection accuracies (%) of each sub-feature in CQP mode

QP	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
F_{fusion}	95.53	96.43	97.64	98.21	97.32	97.32	95.74	94.83	96.43	100	98.21	98.85	95.53	98.21	100	100	96.43
F_{vpar}	85.72	93.04	90.18	90.36	85.71	92.40	84.33	87.05	90.18	91.71	93.04	84.82	86.60	90.82	90.57	91.22	95.71
F_{cpres}	91.25	86.61	87.50	88.39	90.68	93.04	88.57	89.46	95.71	88.39	86.61	87.50	90.46	91.29	89.29	90.18	93.04

QP	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34
F_{fusion}	97.32	98.21	100	97.32	95.53	98.21	96.95	96.42	97.06	97.89	95.43	98.02	99.11	96.13	98.21	95.85	95.64
F_{vpar}	85.71	88.39	95.39	88.39	90.18	91.25	91.82	86.79	92.04	90.61	85.71	90.68	88.39	91.25	85.71	86.61	93.93
F_{cpres}	90.36	84.82	85.71	92.14	88.57	94.82	87.93	85.71	87.50	93.04	85.00	86.61	87.50	84.82	91.25	90.57	91.68

QP	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51
F_{fusion}	96.43	97.33	95.53	97.06	96.52	96.43	98.21	98.21	98.04	100	97.44	94.15	95.88	97.32	100	98.21	97.32
F_{vpar}	90.18	90.18	87.50	86.61	91.07	89.66	88.39	90.18	90.01	93.75	90.80	83.04	83.93	90.18	93.44	89.29	88.39
F_{cpres}	86.61	90.18	91.07	91.96	89.06	90.18	93.75	93.75	92.86	92.86	91.07	86.61	88.39	91.80	91.96	91.96	92.86

Table 6

The detection accuracies (%) achieved by different algorithms.

QP	1	2	3	4	5	6	7	8	9	10	11	12	13
Proposed	95.53	96.43	97.64	98.21	97.32	97.32	95.74	94.83	96.43	100	98.21	98.85	95.53
Jiang's [29]	93.75	95.54	92.86	96.43	93.75	92.87	95.20	90.18	92.86	94.64	94.64	93.75	90.18
Yu's [23]	92.86	93.75	93.75	92.07	95.54	94.64	92.07	93.10	90.32	91.07	95.54	95.54	91.96
Li's [35]	94.64	92.86	90.18	88.62	91.07	95.36	96.43	91.32	87.50	90.72	92.86	93.75	95.53

QP	14	15	16	17	18	19	20	21	22	23	24	25	26
Proposed	98.21	100	100	96.43	97.32	98.21	100	97.32	95.54	98.21	96.95	96.42	97.06
Jiang's [29]	93.75	92.86	95.54	96.43	93.10	96.43	98.54	94.64	96.17	93.10	90.25	92.86	91.96
Yu's [23]	96.43	93.88	95.54	95.54	91.07	93.75	92.86	94.64	85.71	94.64	90.18	93.88	93.88
Li's [35]	87.14	92.86	92.86	90.18	88.39	92.86	93.75	90.27	95.54	96.43	96.05	92.32	93.88

QP	27	28	29	30	31	32	33	34	35	36	37	38	39
Proposed	97.89	95.43	98.02	99.11	96.13	98.21	95.85	95.64	96.43	97.33	95.53	97.06	96.52
Jiang's [29]	97.89	94.64	90.18	93.88	90.80	92.70	98.21	92.86	90.18	92.86	90.18	95.54	90.02
Yu's [23]	90.18	89.29	93.75	90.18	94.64	90.73	94.64	93.75	86.61	93.75	90.18	94.64	93.75
Li's [35]	91.50	93.75	92.86	94.64	94.02	88.39	89.29	90.18	90.79	93.75	95.16	93.75	94.64

QP	40	41	42	43	44	45	46	47	48	49	50	51	Aver.
Proposed	96.43	98.21	98.21	98.04	100	97.44	94.15	95.88	97.32	100	98.21	97.32	97.33
Jiang's [29]	93.70	92.86	95.54	95.54	96.43	90.88	94.50	92.86	92.00	94.64	95.54	95.54	93.77
Yu's [23]	90.18	92.08	87.50	94.64	94.64	91.75	90.18	93.75	92.86	93.07	96.46	97.06	92.83
Li's [35]	88.03	91.96	92.86	87.50	93.75	90.18	90.88	92.86	85.71	91.07	94.64	86.61	92.00

in recompressed videos no matter how QP varies. In the multiple compression process, the coding errors would definitely lead to changes in pixel value. Our proposed detection feature can effectively characterize this type of changes. Therefore, the performance of our method is satisfactory on H.266/VVC double compression detection.

4.4. Performance evaluation in ABR mode

The results in Section 4.3 verify the performance of the proposed algorithm in CQP mode. In this section, performance comparisons are made in ABR mode. Bitrate is respectively set to {2, 3, 4, 5, 6} Mbps by adjusting the option *-b* in VVenC. Note that, QP and rate factor cannot be fixed in this scenario because ABR mode is mutually exclusive with CQP and CRF modes. In order to obtain double compressed H.266/VVC videos, raw YUV sequences are first compressed with bitrate B_1 . Then the generated video clips are decoded into YUV sequences, which are further recompressed with the same bitrate. We define single compressed H.266/VVC video clips as negative samples, while double compressed as posi-

tive samples. We also compare our algorithm with Jiang et al.'s [29], Yu et al.'s [23] and Li et al.'s [35] methods, the results are reported in Table 7: It can be observed from Table 7 that our algorithm performs well on the dataset, averaging 96.79 % when bitrate is selected from {2, 3, 4, 5, 6} Mbps. Our proposed algorithm could achieve higher accuracies in most cases. When bitrate is set to 4 Mbps, the proposed algorithm yields the highest detection accuracy of 98.21%, which is 5.11%, 1.69% and 4.46% higher than those of [29,23,35], respectively. When bitrate is set to 5 Mbps, our algorithm underperforms [23] by 1.01%. The reason can be attributed to the slight change of the inter-frame coding mode since inter prediction modes are exploited in [23], and the simplest Skip mode for MV prediction is used more frequently in this scenario. The algorithm could achieve higher accuracy. On the other hand, it can be seen from these results that the fused feature can achieve more satisfactory results than sub-features in this scenario. This phenomenon infers that the fused feature can achieve promising results, and the subfeatures extracted from the coding process complement each other, which improves the performance of H.266/VVC double compression detection in ABR mode.

Table 7

The detection accuracies (%) in ABR Mode.

Bitrate(Mbps)	2	3	4	5	6	Aver.
F_{fusion}	95.54	96.43	98.21	96.43	97.32	96.79
F_{vpar}	92.40	89.29	90.82	91.25	93.93	91.54
F_{cpre}	91.96	86.61	88.39	94.82	91.18	90.59
Jiang's[29]	94.50	94.64	93.10	93.75	91.96	93.59
Yu's [23]	95.54	93.70	96.52	97.44	93.70	95.38
Li's[35]	95.43	92.83	93.75	92.83	90.18	93.00

Table 8

The detection accuracies (%) with different GOP sizes.

GOP		QP					Aver.
		10	20	30	40	50	
10	Proposed	100	100	99.11	96.43	98.21	98.75
	Jiang's[29]	94.64	98.54	93.88	93.70	95.54	95.26
	Yu's [23]	91.07	92.86	90.18	90.18	96.46	92.15
	Li's[35]	90.72	93.75	94.64	88.03	94.64	92.36
	Proposed	97.25	95.24	96.43	95.24	97.32	96.30
20	Jiang's[29]	92.86	95.24	90.18	92.86	93.75	92.98
	Yu's [23]	90.18	90.18	88.39	90.83	93.75	90.67
	Li's[35]	88.39	93.75	93.75	87.50	93.75	91.43
	Proposed	92.86	93.75	94.64	91.07	92.86	93.04
	Jiang's[29]	90.18	91.07	88.39	90.18	89.29	89.82
30	Yu's [23]	90.18	87.50	86.61	87.50	90.18	88.39
	Li's[35]	86.61	91.17	90.18	86.61	89.29	88.77
	Proposed	89.29	90.18	92.86	91.07	87.50	88.18
	Jiang's[29]	85.71	88.39	84.82	86.61	87.50	86.61
	Yu's [23]	85.71	86.61	84.82	85.71	88.39	86.25
40	Li's[35]	84.82	89.29	86.61	85.71	83.04	85.89

4.5. Performance evaluation with different GOP sizes

In the experiments described in Section 4.2 4.3 4.4, the GOP size is fixed at 10. However, in the real coding scenario, the size could be adjusted according to actual needs. Therefore, it is of great significance to evaluate the performance of the proposed algorithm when videos are coded with different GOP sizes. Since VVenC is used as the encoder, the GOP size can be modified by adjusting the value of -g in the command line. In this experiment, GOP size is selected from {10, 20, 30, 40}, and we verify the performance when QP is set to 10, 20, 30, 40, 50 respectively. By extracting CU partition and prediction modes during the decoding procedure, the feature can be constructed. Following the steps described in Section 3.5, results are shown in Table 8.

We can observe from Table 8 that no matter how the GOP size varies, the detection accuracy of our method is generally above 87%. The proposed algorithm achieves the highest accuracy of 100% in the cases of $GOP = 10$, $QP = \{10, 20\}$, which means all the sequences are correctly classified. With the increase of GOP size, the accuracies of these methods would decrease, but the proposed method maintains relatively higher values. This phenomenon can be attributed to the corresponding decrease in the number of I-frames as the GOP increases. Features in the method are mainly based on the information in I-frames. The increase of GOP size results in the insufficiency of traces in I-frames that the algorithm relies on. The proposed algorithm also performs well when $GOP = 20$, averaging 96.30 %, which is 3.32%, 5.63%, and 4.87% higher than those of [29,23,35], respectively. The main reason is that the coding modes including intra prediction and partition modes in H.266/VVC is somewhat different from H.265/HEVC or the previous coding standards. By exploring and constructing features compatible with the coding standard, the algorithm can achieve better performance.

4.6. Performance evaluation with different video spatial resolution

In this section, we further evaluate the performance of our algorithm when 4 K resolution ultra-high definition (UHD) videos are selected as the test samples. 15 4 K raw YUV sequences from SJTU-4 K Dataset [36] are downloaded from public website [37] and their names are listed in the footnote below². All these sequences are larger than 200 GB in size, with YUV 4:2:0 color sampling, 8 bits per sample formats and 300 frames. The factors such as image texture, image detail, the movement speed of the object in the image, light intensity, and the camera lens stretching, panning are taken into account [36]. All these sequences are also split into clips with a length of 100 frames to increase the sample quantity, then a total of 45 raw YUV clips can be obtained. In this experiment, VVenC and VTM are used to generate single and double compressed H.266/VVC video clips. QP is selected from {1,2,...,51}. Thus, a total of 4590 video sequences can be obtained. By performing the detection, results can be obtained, as shown in Fig. 10.

Data in Fig. 10 shows that the proposed algorithm achieves an average accuracy of 96.89%, which is 2.99, 3.54, and 4.76 percent higher than those of [29,23,35] respectively. When QP is set to {5,16,33}, our algorithm achieves the highest values, and 46 of these 51 values are higher than the other three comparison algorithms. The results suggest that our algorithm could perform better than existing algorithms in H.266/VVC double compression detection on 4 K resolution ultra-high definition (UHD) video set. This finding is understandable because our algorithm is based on the analysis of CU partition and prediction modes. We use MUM and SPM strategies to facilitate the feature calculation, and the varia-

² The names of the YUV sequences in SJTU-4 K dataset: *Bund Nightscape, Campfire Party, Construction Field, Library, Fountains, Marathon, Residential Building, Runners, Rush Hour, Tall Buildings, Traffic and Building, Traffic Flow, Tree Shade, Scarf, Wood.*

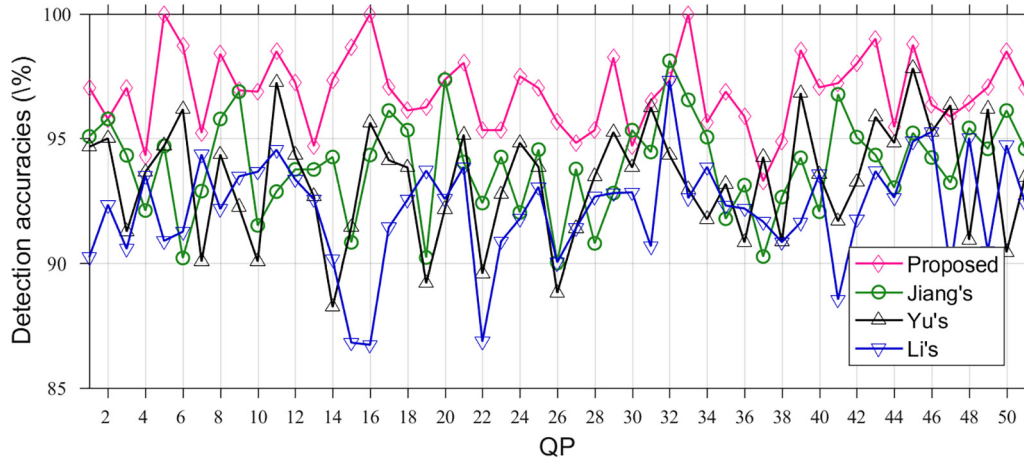


Fig. 10. Performance evaluation on SJTU-4 K Dataset.

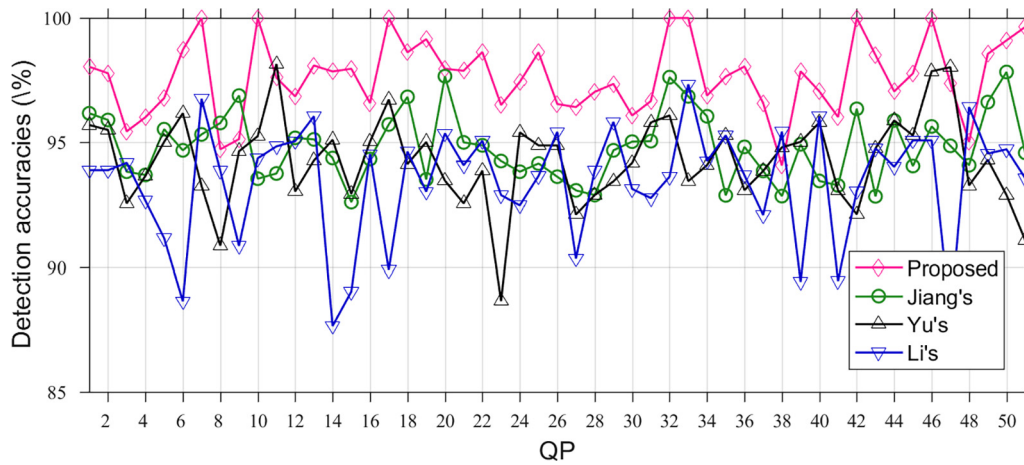


Fig. 11. The detection accuracies (%) of multiple compressions detection.

tion of partition modes, together with the consistency of prediction modes are calculated for feature construction. Even though the increase of video resolution results in higher time consumption of feature extraction, it would not affect the characterization of recoding traces. In summary, these results clearly demonstrate the robustness of the proposed algorithm for H.266/VVC double compression detection.

4.7. Performance evaluation of multiple compressions detection

Multiple compressions of video often occur in the real scenario. In view of this, it is of great significance to evaluate the sensitivity of our algorithm to multiple compressions. In this experiment, we first compress the raw YUV sequences to generate a single compressed H.266/VVC videos. Then these videos are decoded and further encoded to achieve video multiple compressions. The output of each decoding is used as the input of the next coding. We repeat this operation 3 times to generate multiple compressed videos. In this experiment, the parameters of each compression are the same, and multiple compressed videos are defined as positive samples, while single compressed videos are labeled as negative samples. We traverse QP from 1 to 51, and the results under each QP are shown in Fig. 11.

As can be observed from Fig. 11, the detection accuracies of our algorithm are still relatively higher than the comparison algorithms when positive samples are multiple compressed H.266/VVC video clips. Compared with the results of double compression detection in Table 6, these algorithms can achieve better results. For example, the average accuracies of our algorithm and [29,23,35] respectively reach 97.64%, 94.82%, 94.29%, 93.44%, which are higher than the values in Table 6. The better performance can be attributed to the fact that multiple compressions would introduce more obvious compression artifacts, making the features more distinguishable. As shown in Fig. 12, we perform consecutive encoding and decoding on *tractor.yuv*. Fig. 12(a) is the content of the first frame of the sequence; Figs. 12 are the partially enlarged contents of the first frames of the single, double and triple compressed videos. It can be observed that with the compression time increases, the contents of the same frame of the videos are basically the same, and the differences cannot be detected by the naked eye. However, by analyzing their respective CU partition modes in Figs. 12(e), (f) and (g), it can be found that the partition modes at the same location of the frames may change due to recompression, and this variation is more likely to occur in areas with complex textures. Besides, as the compression time increases, the difference of partition modes between triple and single compressed videos are more significant than that between dou-

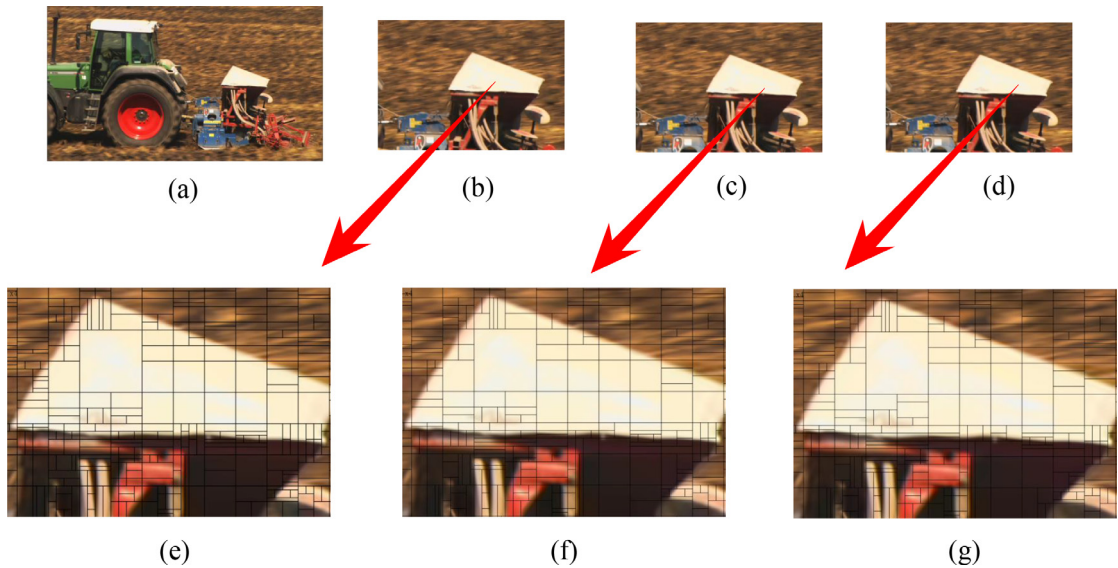


Fig. 12. (a) The content of the first frame of *tractor.yuv*. (b–d) The partially enlarged contents after single, double and triple compressions. (e–g) The partition modes of the partially enlarged contents in (b), (c) and (d).

ble and single compressed videos. Therefore, better performance can be achieved in this scenario.

4.8. Sensitivity to different -Preset options

Similar to the encoders of the previous coding standards, e.g., X265, X264, the H.266/VVC encoder also supports different -Preset options to optimize the trade-off between encoding speed and coding efficiency. Since VVenC is used as the encoder in this work, we learned that vvenccapp of VVenC can be used in one of five predefined presets. In the slowest preset, the encoder reaches the highest compression gain, whilst in the fastest preset the runtime is significantly decreased [38]. To verify the effectiveness and robustness of the proposed algorithm when different -Preset options are used, we performed related experiments on the dataset introduced in Section 4.1. In this experiment, the option -Preset is selected from {faster, fast, medium, slow, slower}. Bitrate is respectively set to {2, 3, 4, 5, 6} Mbps. Other parameter settings are

described in Section 4.1. Note that, the coding parameters of single and double compressed videos are consistent. We also compare our algorithm with [29,23,35], detection accuracies of these four methods are reported in Table 9.

From Table 9, it can be found that the -Preset option does have an impact on the performance of the algorithms. Taking the average accuracy as an example, when -Preset is set to slower, the average accuracies of the algorithms reach the highest values, which are 98.61%, 95.83%, 96.85%, 94.41%, respectively. However, the methods yield the lowest accuracies of 94.24%, 89.54%, 91.96% and 88.58% when -Preset is set to faster. It is because when faster preset is adopted, the encoder takes shortcuts to improve performance at the expense of quality and compression efficiency. If a slower preset is used, the encoder will use more computations to achieve the best quality at the selected bit rate, and more detailed information is preserved in this case. It also can be observed that the proposed algorithm is superior to the other three algorithms with different -Preset options. For example, when -Preset is set to

Table 9
The detection accuracies (%) with different -Preset options.

		Bitrate					Aver.
Preset		2	3	4	5	6	
faster	Proposed	95.54	93.04	94.23	93.75	94.64	94.24
	Jiang's[29]						
	Yu's [23]	91.07	90.18	87.50	90.18	88.77	89.54
	Li's[35]	91.07	91.07	94.64	92.86	90.18	91.96
	Li's[35]	92.86	89.29	85.71	88.77	86.25	88.58
fast	Proposed	93.04	93.75	95.24	95.24	96.43	94.74
	Jiang's[29]						
	Yu's [23]	92.86	92.86	91.07	90.18	91.07	91.61
	Yu's [23]	93.75	90.18	95.54	94.64	92.83	93.39
	Li's[35]	93.75	93.75	90.18	89.29	90.18	91.43
medium	Proposed	95.54	96.43	98.21	96.43	97.32	96.79
	Jiang's[29]						
	Yu's [23]	94.50	94.64	93.10	93.75	91.96	93.59
	Yu's [23]	95.54	93.70	96.52	97.44	93.70	95.38
	Li's[35]	95.43	92.83	93.75	92.83	90.18	93.00
slow	Proposed	96.43	96.43	100	98.21	97.32	97.68
	Jiang's[29]						
	Yu's [23]	95.54	93.75	95.54	96.43	92.86	94.82
	Yu's [23]	94.82	94.82	97.32	97.44	95.54	95.99
	Li's[35]	96.43	91.17	94.82	93.04	89.29	92.95
slower	Proposed	98.21	100	100	96.43	98.40	98.61
	Jiang's[29]						
	Yu's [23]	96.43	95.24	96.43	97.32	93.75	95.83
	Yu's [23]	95.24	95.24	97.32	100	96.43	96.85
	Li's[35]	95.24	96.43	96.43	93.75	90.18	94.41

slow, an average accuracy of 97.68% is achieved by our algorithm, that is + 2.86%, +1.67% and + 4.73% higher than those of [29,23,35]. This comes from the fact that a series of new coding features are introduced in H.266/VVC. By leveraging CU partition and prediction modes analysis, the proposed detection features can effectively represent the recompression trace, which lead to the better performance of our algorithm.

5. Conclusion

In this paper, a novel algorithm is proposed for the detection of H.266/VVC double compression with the same coding parameters. Based on the in-depth analysis of the generation of H.266/VVC double compression traces, we find that the quality degradation characteristic still exists in H.266/VVC double compression. Then, the variation of CU partition and consistency of prediction modes are exploited to characterize double compression traces. In order to reduce the interference of quadtree or multi-type tree partition to mode analysis, and also to facilitate feature construction, we define Minimum Unit Mapping (MUM) and Subunit Prediction Mapping (SPM). We calculate the strength of the variation in the number of CUs and the consistency ratio of adjacent prediction mode pairs in the horizontal, vertical, major diagonal, and minor diagonal directions after multiple compressions on the basis of the mapped data. Further, CU partition and prediction modes based features are concatenated into a 60-D vector for the detection. Extensive experimental results demonstrate the better performance of the proposed algorithm. Meanwhile, the results also confirm that the algorithm is robust in different coding scenarios.

As mentioned in Section 1, the H.266/VVC standard introduces a series of new techniques, which makes the coding decision strategies to be more complex than those of previous standards. Therefore, it is still an open problem to make a thorough analysis on the difference of the trace information between single and double compressed H.266/VVC videos. In future works, we will further explore the detection of transcoded H.266/VVC videos and analyze the traces left by video transcoding. Moreover, we will also try to exploit CUR decomposition and deep learning based methods for the processing of multimedia information.

CRediT authorship contribution statement

Qiang Xu: Conceptualization, Methodology, Writing - original draft. **Dongmei Xu:** Writing - review & editing, Validation, Visualization. **Hao Wang:** Software. **Zhongjie Mi:** Data curation. **Zhe Wang:** Investigation. **Hong Yan:** Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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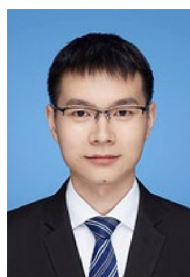
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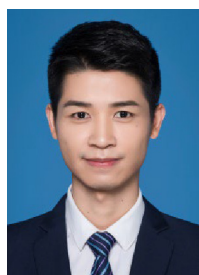
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